Tobias Mwangi-Hartley

Data Scientist | Machine Learning Engineering Focus
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PROFESSIONAL SUMMARY

Data scientist with six years in analytics and data science, including over four years developing forecasting, churn and experimentation solutions for a subscription business. Strong Python, feature engineering and statistical evaluation, with operational batch workflows using BigQuery and Vertex AI Workbench. Moving into machine learning engineering, bringing measurable business impact, automated testing experience and collaboration with product, growth and data platform teams.

EXPERIENCE

Data Scientist | Thistlebox, meal-kit subscriptions
Feb 2022 - Present

- Built a LightGBM demand forecasting model predicting weekly orders by recipe and region six weeks ahead, with calendar, price and menu-position features. Reduced WAPE from 23% to 14%, with an estimated 9% reduction in food waste.
- Run forecasting as a Vertex AI Workbench scheduled notebook, writing outputs to BigQuery for operations to access through Looker.
- Introduced pytest unit tests for forecasting feature code in 2025; coverage is approximately 60%.
- Developed an XGBoost churn model with weekly batch scoring in BigQuery to target retention offers. Evaluation using a 10% holdout showed churn 2.1 percentage points lower in the treated group.
- Write production SQL and maintain dbt models for the analytics layer; use dbt Cloud scheduling and scheduled notebooks.
- Built a Streamlit sequential A/B testing calculator used by approximately 25 product and growth colleagues. Wrote the Dockerfile; the data platform team handles deployment on Cloud Run.
- Design and analyse approximately 40 A/B tests annually, present results to leadership monthly and delivered internal "Stats for PMs" training three times.

Junior Data Analyst | Firth Energy Retail
Sep 2020 - Jan 2022

- Performed customer segmentation and tariff analysis using R and SQL.
- Automated monthly regulatory reports with R Markdown, saving the team approximately two days each month.

TECHNICAL SKILLS

- Python: pandas, scikit-learn, LightGBM, XGBoost, statsmodels, SHAP; pytest unit testing.
- Data processing and cloud tools: SQL, BigQuery, dbt, dbt Cloud scheduler, Vertex AI Workbench, Looker.
- Engineering tools: Git version control, Jupyter, Streamlit; Dockerfile authoring for a Streamlit app.
- Modelling and evaluation: feature engineering, time-series forecasting, A/B testing, causal inference, Bayesian statistics; R, tidyverse, brms.

SELECTED LEARNING PROJECT

Image Classification | Personal course project | 2025

- Fine-tuned a small CNN in PyTorch on a public food-images dataset, achieving approximately 80% accuracy. Used torchvision and Google Colab; a learning project, not a production deployment.

EDUCATION

MSc Statistics, Distinction | University of the Lothians | 2020
Dissertation: Bayesian hierarchical models for small-area energy demand

BSc Mathematics | University of the Lothians | 2019

LANGUAGES

English: native | Swahili: conversational